Photovoltaic Devices

The following subjects will be covered

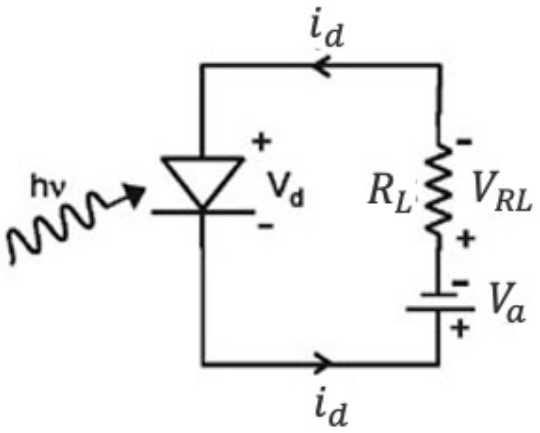
- Photovoltaic Mode of Operation.
- Basic Solar Cell Structure.
- Solar Radiation Fundamentals.

Photo-voltaic

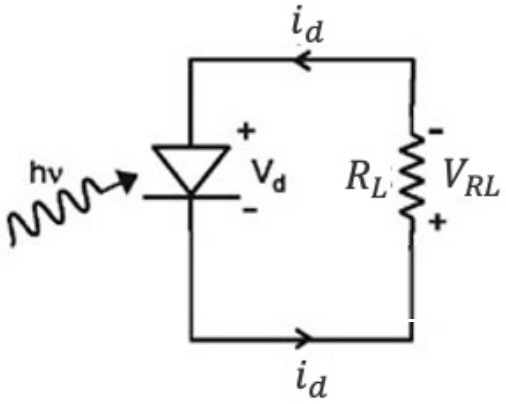


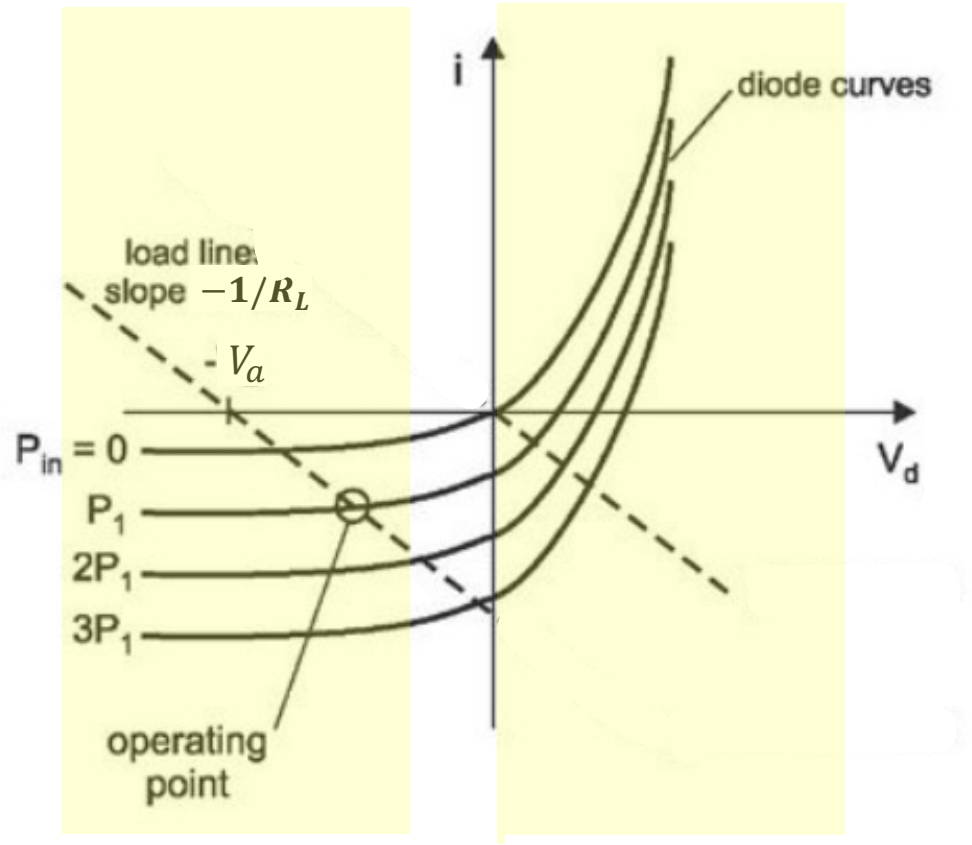
Photodiode Modes of Operation

Photoconductive Mode



Photovoltaic Mode

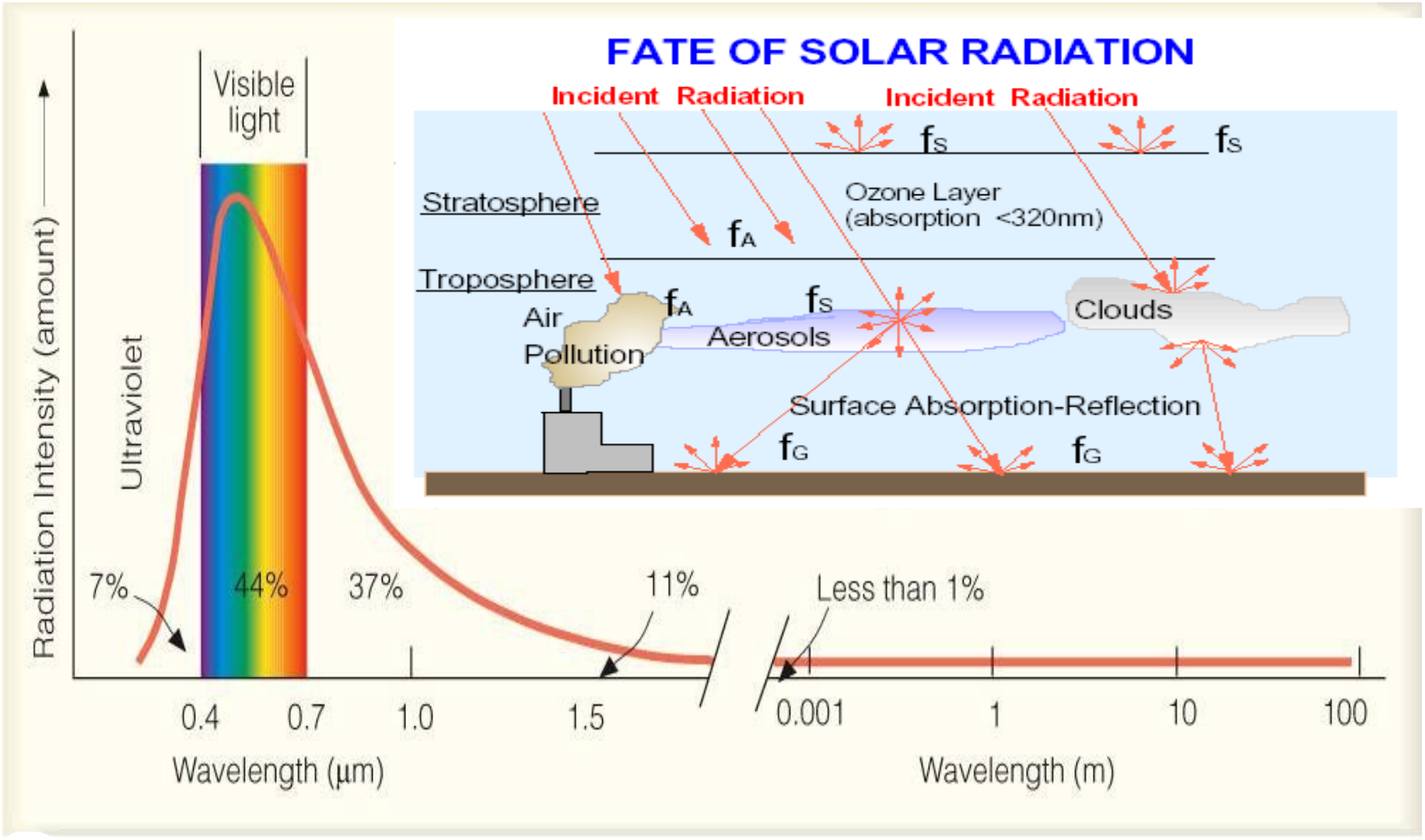




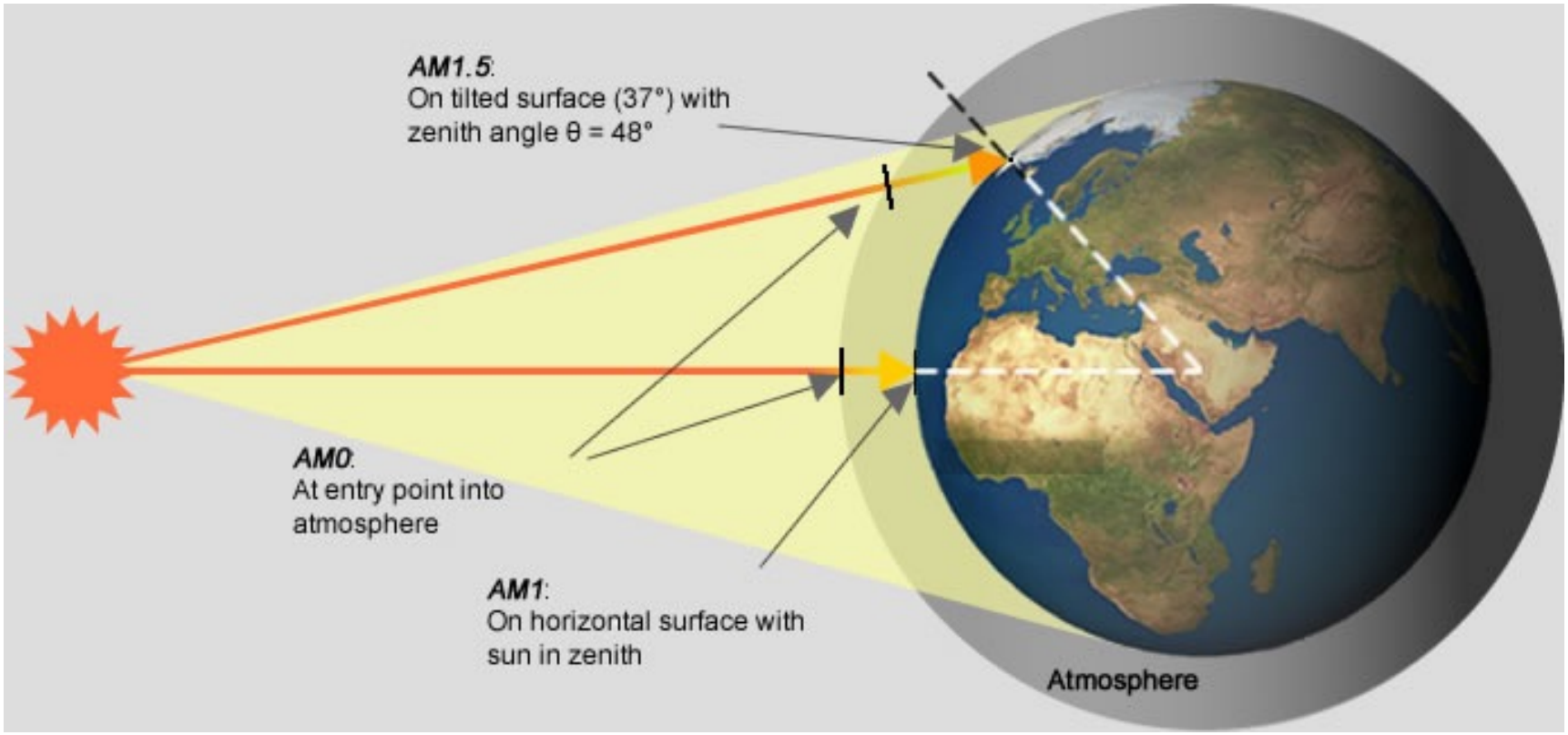
Photoconductive Mode

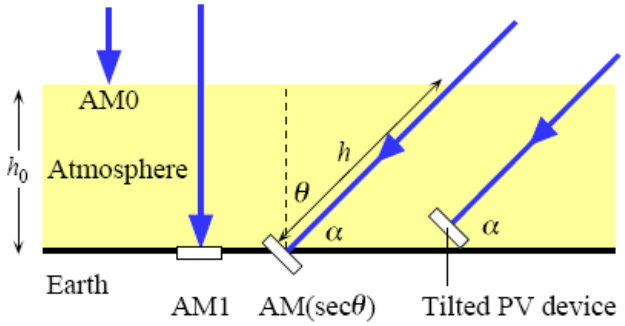
Photovoltaic Mode

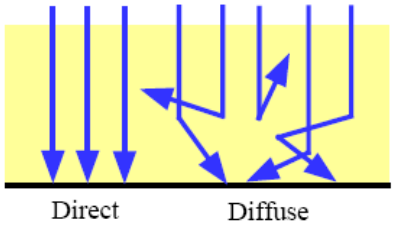
Solar Spectrum



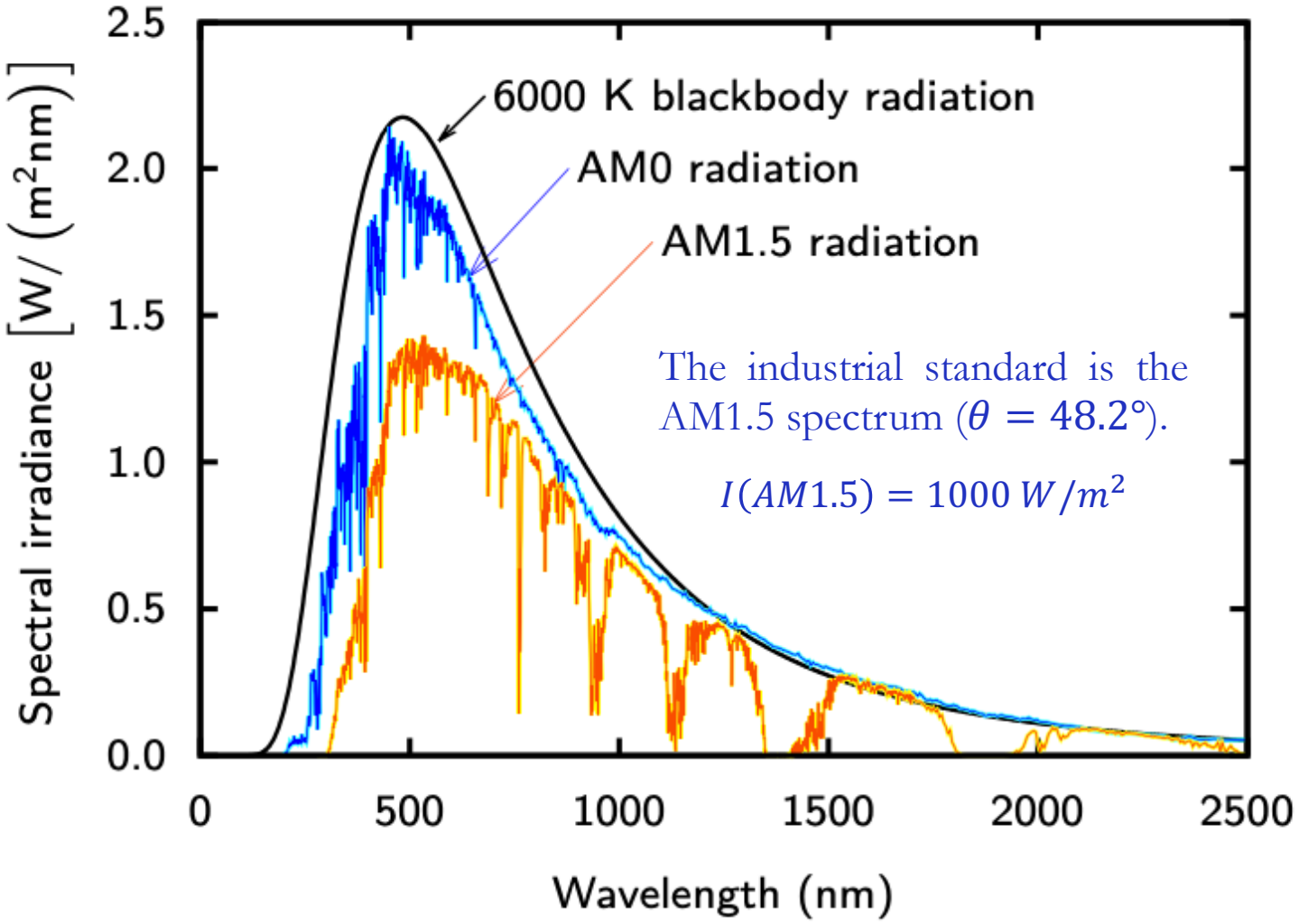
Solar Spectrum



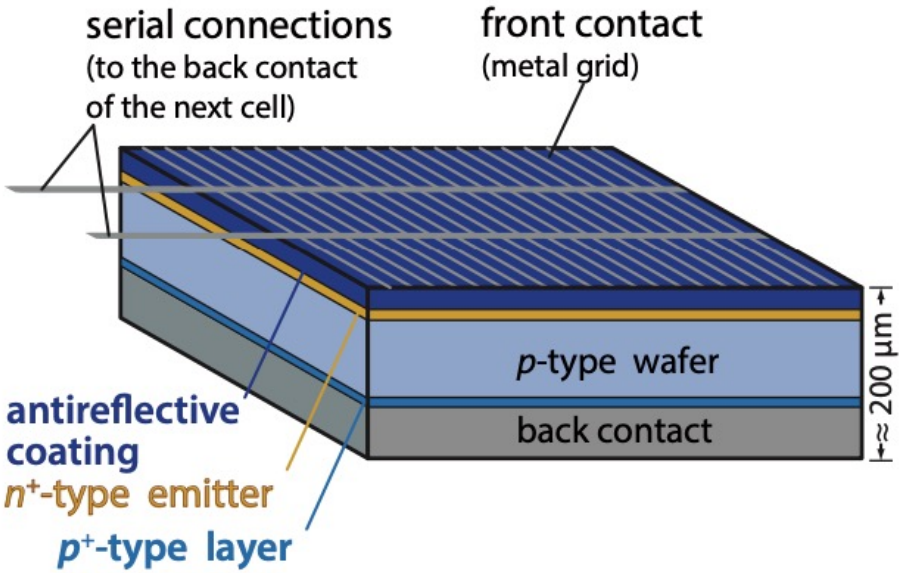


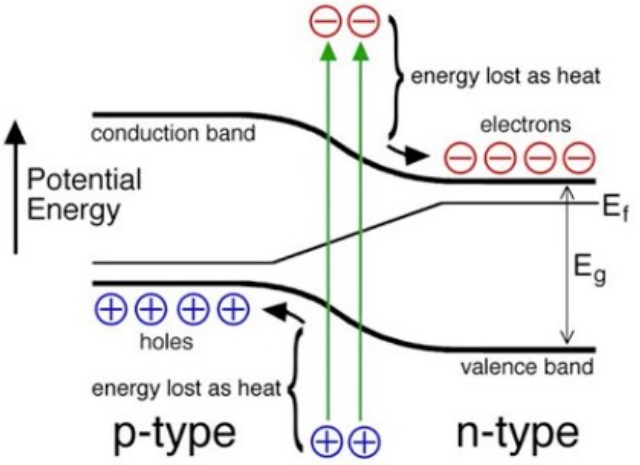


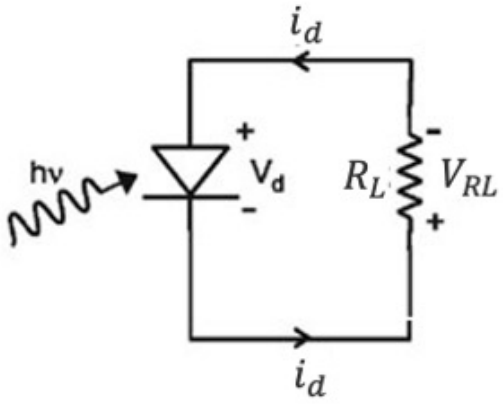
Solar Spectrum



Principle of Operation of PN Solar Cells







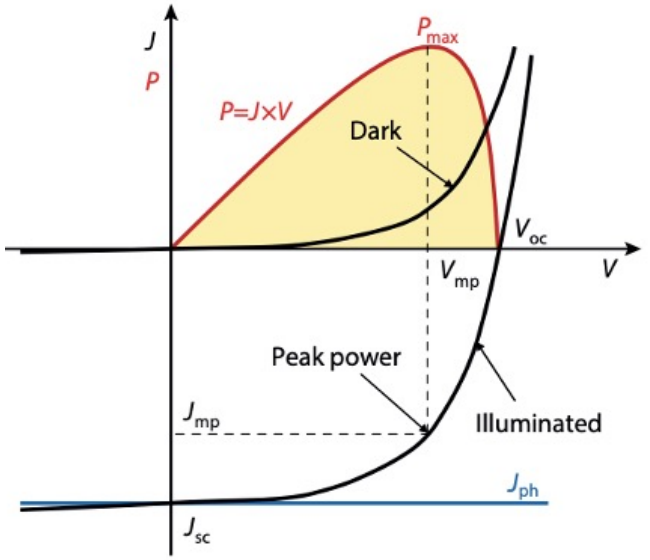
The conversion efficiency is characterized by:

1) Open-circuit voltage.

V_{oc}

2) Short-circuit current

I_{sc}



Design Aspects: Losses and Efficiency

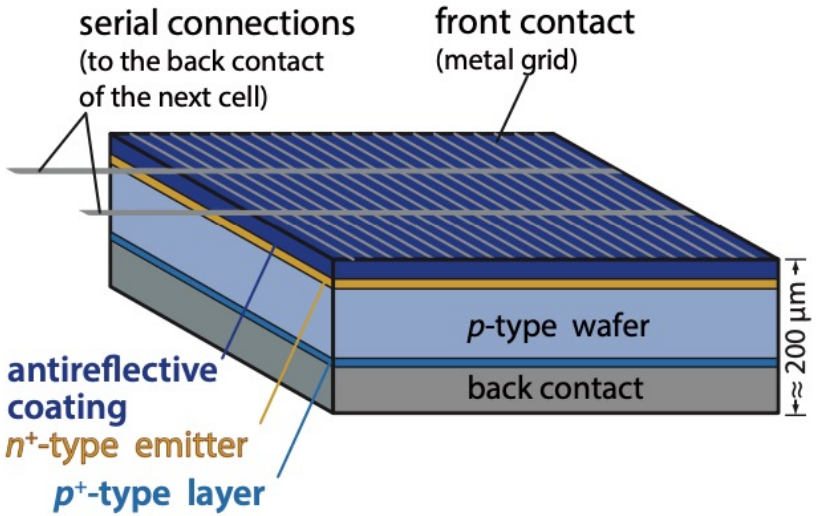
Factors affecting Open-circuit voltage V_{oc} and Short-circuit current I_{sc}.

Recombination.

Bulk Recombination because of defects and impurities. Also because of short minority lifetime.

Surface Recombination because of surface roughness and surface states (dangling bonds).

Fabrication process needs to be perfected.



Design Aspects: Losses and Efficiency

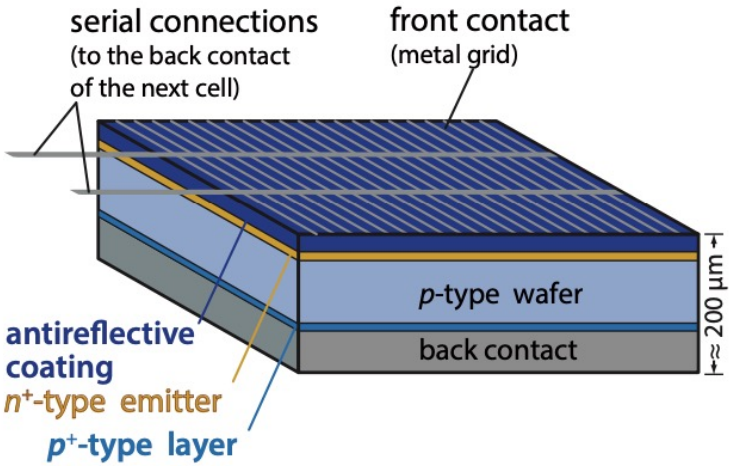
Factors affecting Open-circuit voltage V_{oc} and Short-circuit current I_{sc}.

Reflection and Transmission-through.

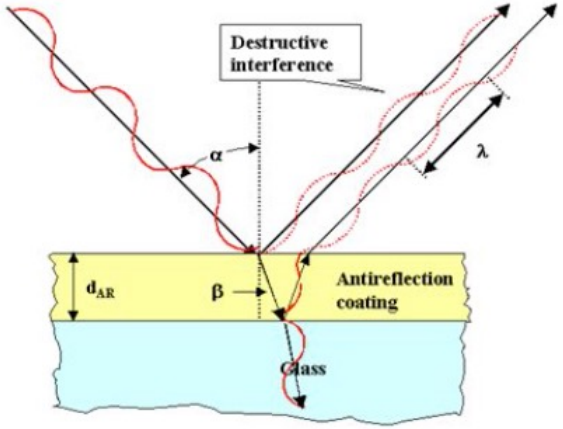
Free area ratio. $\frac{A_{free}}{A_{tot}}$

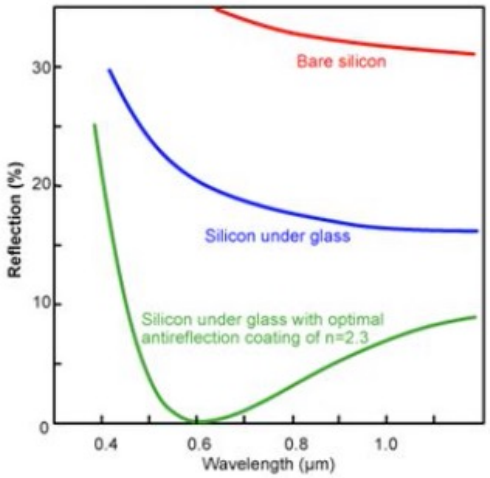
Fresnel Reflection. $(1 - R)$

Transmission-through. Device thickness.



- Electrode grid management.
- Anti-reflection coating.
- Back mirror.
- Surface texturing.





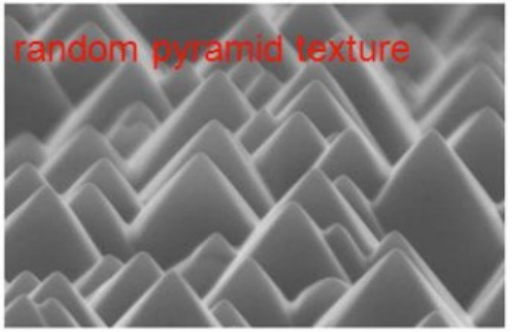
Design Aspects: Losses and Efficiency

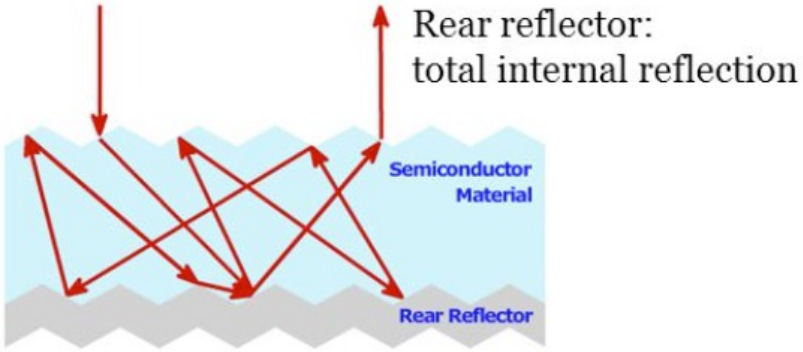
Factors affecting Open-circuit voltage V_{oc} and Short-circuit current I_{sc}.

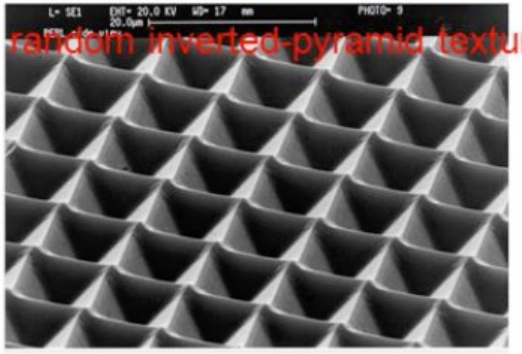
Reflection and Transmission-through.

Surface Texturing.









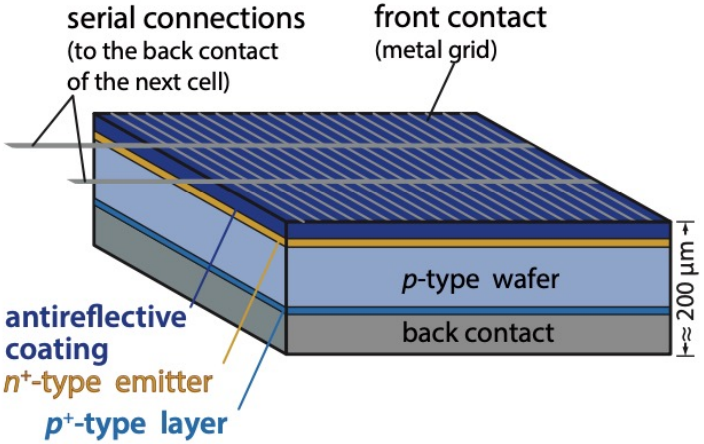
Design Aspects: Losses and Efficiency

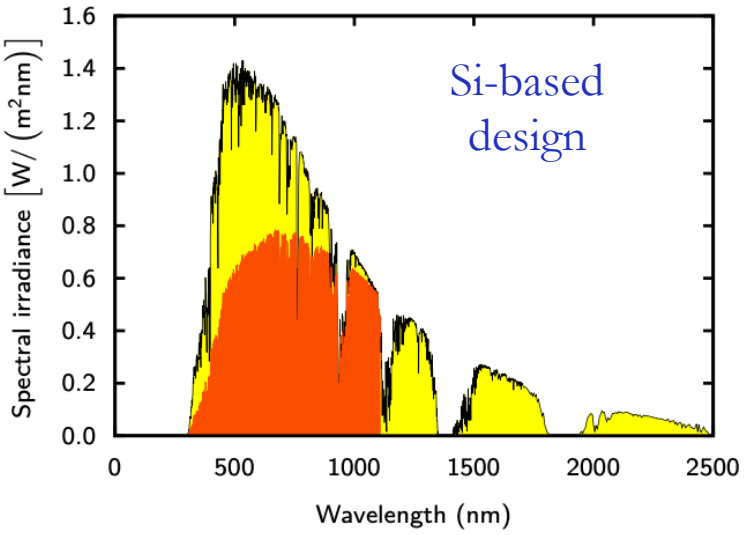
Factors affecting Open-circuit voltage V_{oc} and Short-circuit current I_{sc}.

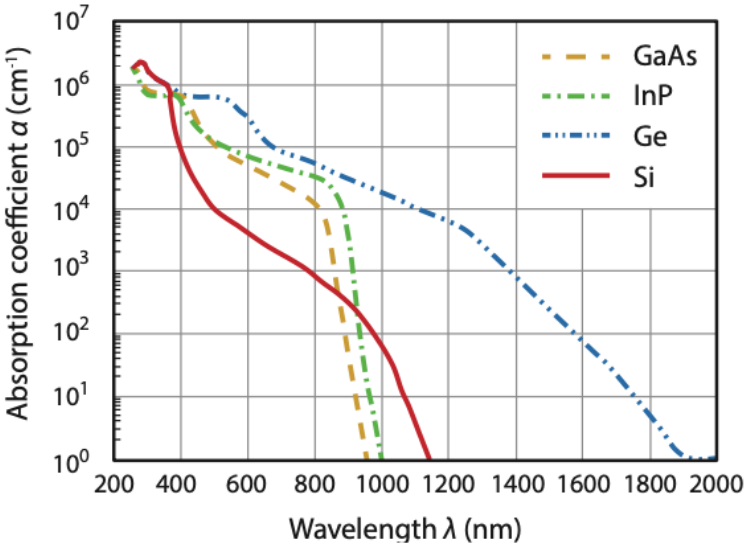
Partial Spectrum Absorption.

Limited material absorption (cutoff wavelength)

- Multi-material (bandgap) design.





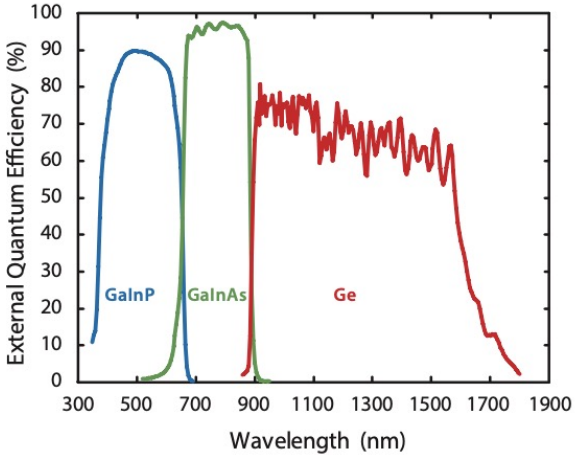


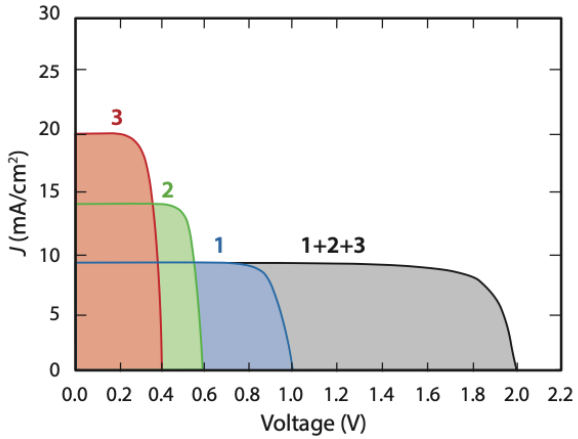
Design Aspects: Losses and Efficiency

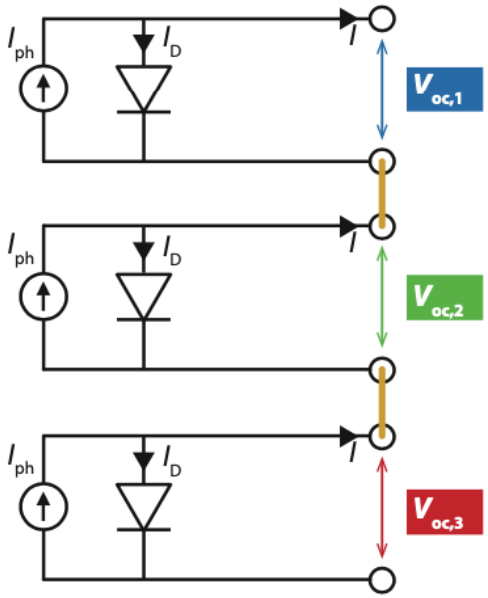
Factors affecting Open-circuit voltage V_{oc} and Short-circuit current I_{sc}.

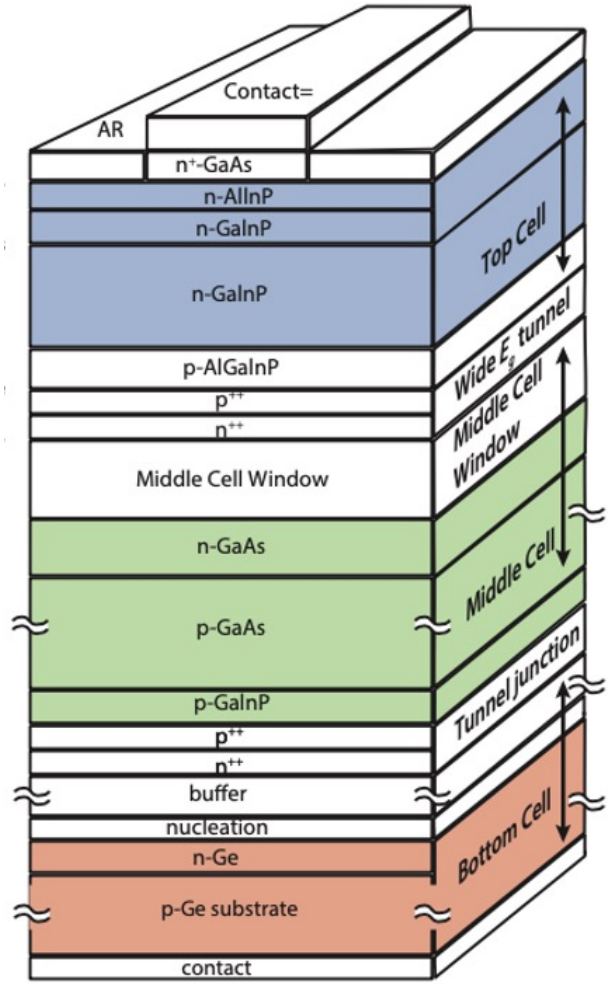
Partial Spectrum Absorption.

Multi-material (bandgap) design









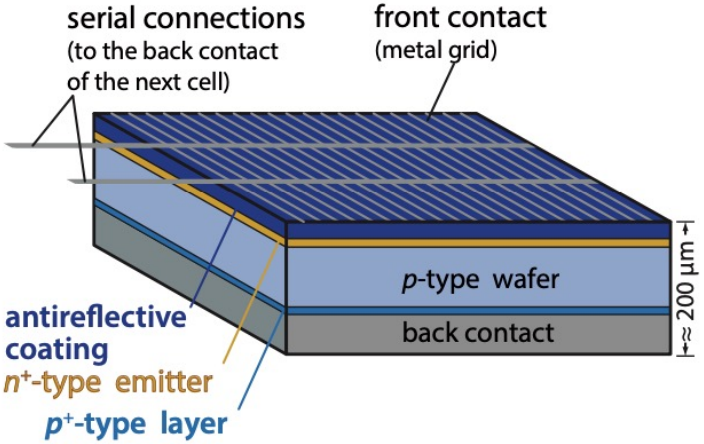
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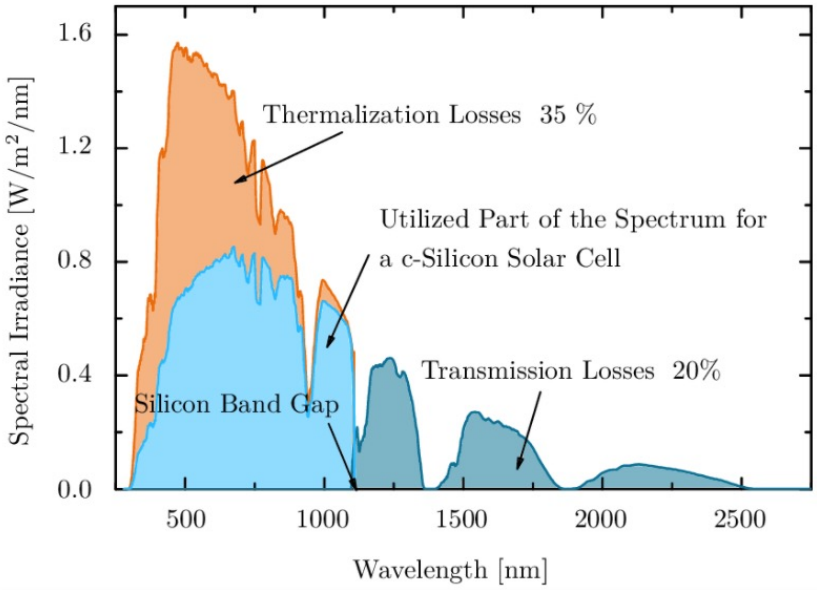
Factors affecting Open-circuit voltage V_{oc} and Short-circuit current I_{sc}.

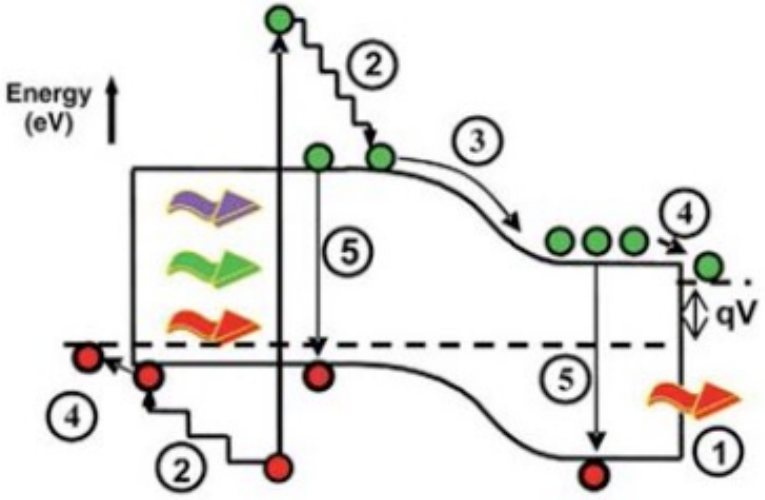
Thermalization.

Extra photon energy in converted to heat ($\hbar\omega > E_g$).

- Multi-material (bandgap) design.



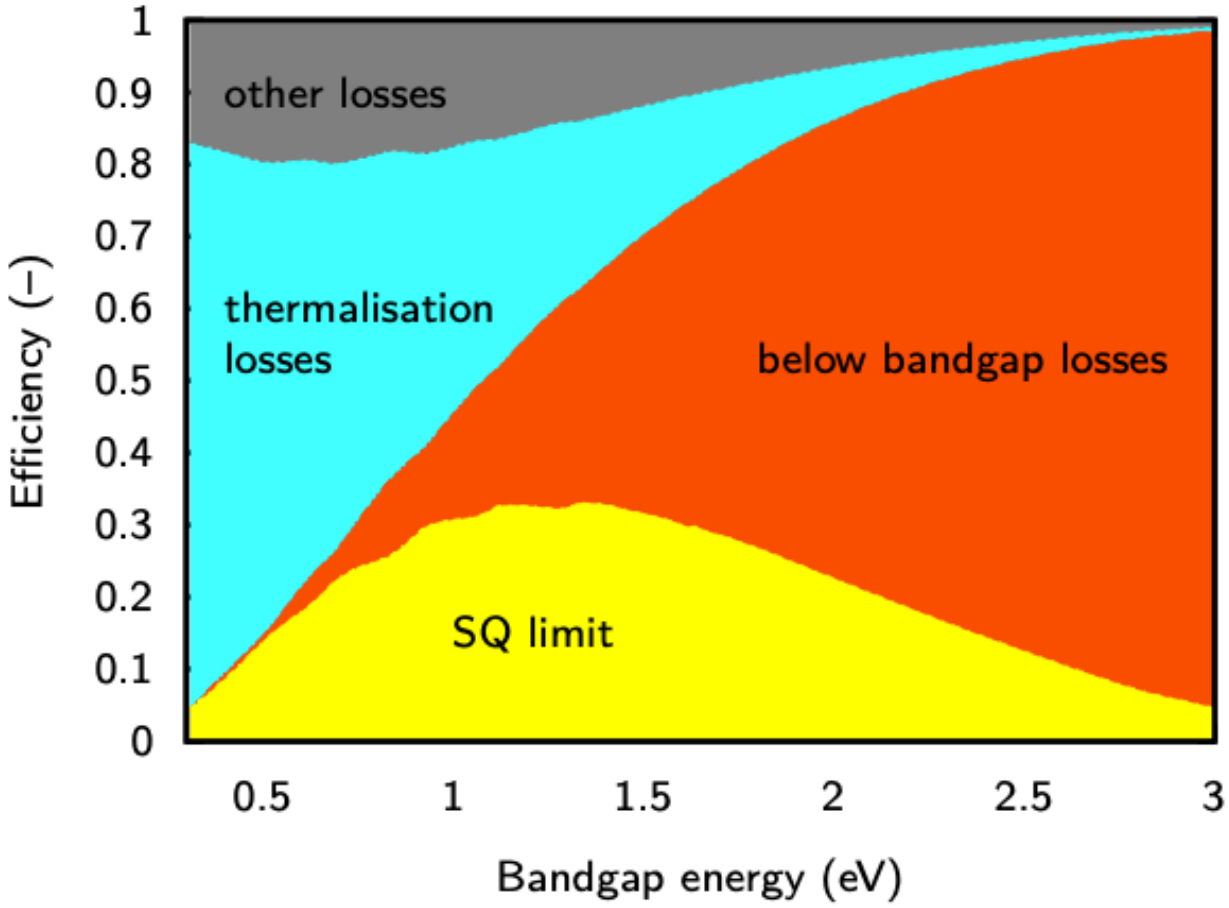




Design Aspects: Losses and Efficiency

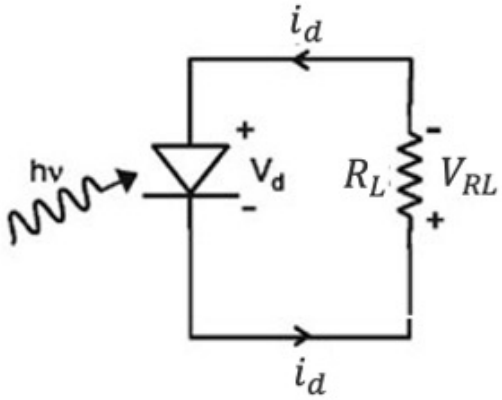
Factors affecting Open-circuit voltage V_{oc} and Short-circuit current I_{sc}.

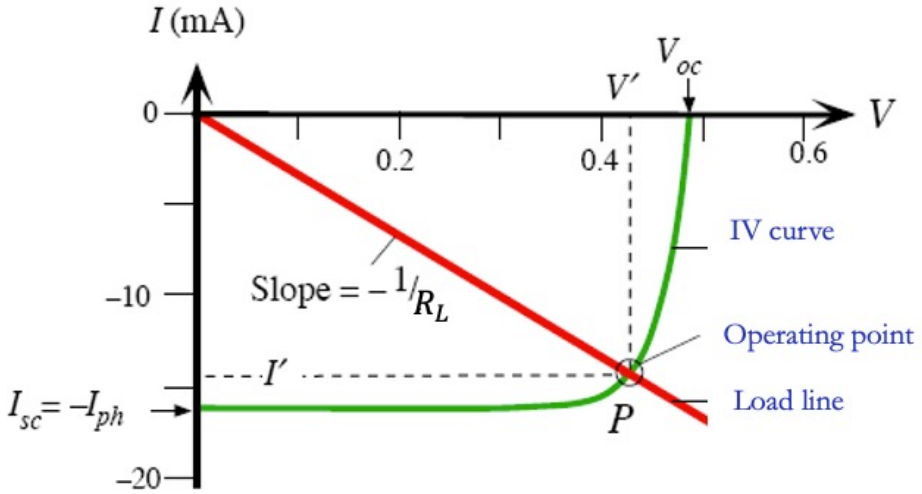
Power Budget



Solar Cell Circuit Analysis

To find the operating point, we draw the circuit load line.





Open-circuit voltage, V_{oc} , is the maximum voltage the cell can produce with no load. It is given by

$$V_{oc} = \frac{kT}{q}\ln(1 + i_p/i_s).$$

Short-circuit current, I_{sc}, is the maximum current the cell can produce when short-circuited. The small series resistance can reduce it.

The fill factor is defined as

$$FF = \frac{P_{cell}}{P_{max}} = \frac{I'\,V'}{I_{sc}\,V_{oc}}$$

The conversion efficiency

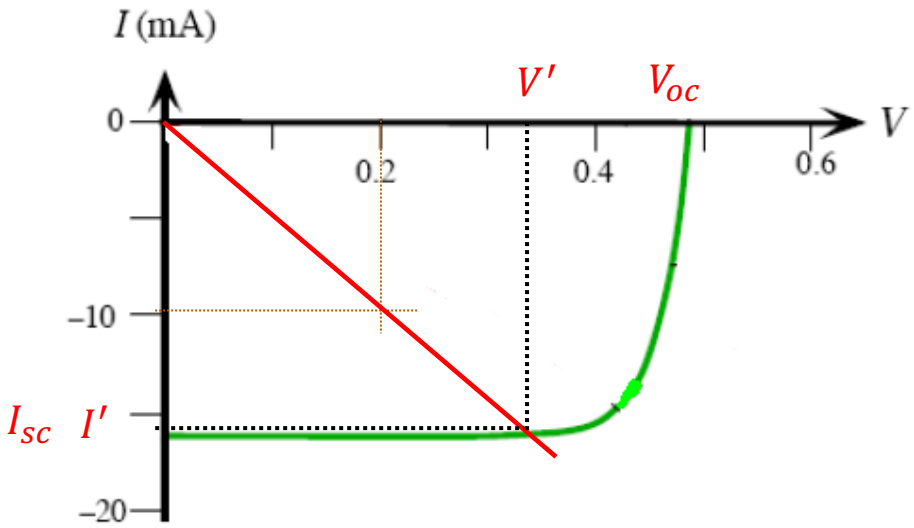
$$\eta = \frac{P_{cell}}{P_{in}} = \frac{I'\,V'}{P_{in}} = \frac{FF\,I_{sc}\,V_{oc}}{P_{in}}$$

Example

The IV curve of a certain Si solar cell with a total cell area of 4 cm^2 is given in the figure.

The cell is under illumination of 600 Wm^{-2}, and is connected to drive a load of R=20 Ω.

Calculate the fill factor and the efficiency of this solar cell?



$$FF = \frac{(16\,mA)\,(0.33)}{(16\,mA)(0.48)} = 69\,\%$$

$$\eta = \frac{(16\,mA)\,(0.33)}{(600\,W/m^2)(4\times10^{-4}m^2)} = 2.2\,\%$$

What should the load be to obtain maximum power transfer from the solar cell to the load?

Q & A

Send me an email if you have any questions.